



SECOND SEMESTER SESSION 2025/2026 (A252)

FINAL REPORT

**CUSTOMER CHURN PREDICTION IN THE TELECOMMUNICATIONS
INDUSTRY USING DATA MINING TECHNIQUES**

Telco Churn Prediction System Link:

<https://telco-churn-prediction-mly57bjbcpgnh48swfsz9a.streamlit.app/>

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ABSTRACT

Customer churn refers to when a subscriber stops using a service and switches to a competitor. Keeping a current customer is much cheaper than getting a new one. Studies show that acquiring a new customer can cost 5 to 25 times more than retaining an existing one. Most existing prediction tools require complex coding skills, which makes them difficult for business managers to use. There is also no clear agreement in the literature on which machine learning model works best for telecom churn data. This study uses the Knowledge Discovery in Databases (KDD) process to build, compare, and deploy churn prediction models. The dataset used is the Telco Customer Churn dataset, which contains 7,043 customer records and 21 attributes. The number of input variables was reduced from 21 to 8 based on findings from the literature. Six models were trained and tested using a 70:30 data split. These models are of four types of Decision Trees, one Random Forest and one Logistic Regression. For each model, the accuracy, the precision, the recall, and the F1-Score were obtained. Logistic Regression performed best in terms of accuracy (80%), precision (67%), recall (53%) and F1 score (59%). Contract Type, Tenure and Monthly Charges were the top three churn predictors identified. The best model was then deployed in a Streamlit web dashboard. This dashboard allows non-technical users to enter customer details and get a real-time churn risk result, along with a chart showing which factors influenced the prediction and a suggested action.

Keywords: Customer Churn Prediction, Telecommunications, Data Mining, KDD Process, Logistic Regression, Decision Tree, Python, Streamlit.

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CHAPTER 1: INTRODUCTION

1.1 Background of Study

Telecommunications companies are losing customers at a high rate. It's known as customer churn. It's what occurs when a person cancels his subscription with one service and subscribes to another. It's significantly cheaper to keep customers than to acquire new ones. Studies have revealed that it costs 5-25 times more to acquire a new customer than to retain an existing one (Imani, 2024; Banarase et al., 2024). This puts churn as a real financial issue, rather than just a marketing issue.

The telecom industry is different to the retail industry. A customer in a retail business can go to several stores at a time. Typically, a customer remains with one provider for telecom. The loss is irrevocable and permanent when they depart. Customers can easily switch, especially if they are on a short contract, due to competition within the market (Saeed, 2025).

There is a lot of data out there for customers already collected by many companies. What they don't have is a simple tool they could use that data to figure out who will leave. Current solutions are heavily reliant on programming, which is not something business managers are able to do. In this study, a simple-to-use, non-coding prediction system is built.

1.2 Problem Statement

There are three major issues with customer attrition for telecom companies. High churn rates are the first issue. No penalty for customers who are on a monthly contract. This makes them the most likely group to churn (Imani, 2024; Saeed, 2025). It is difficult for companies to retain these customers because there is not a lock-in period.

The second problem is the black box problem. A lot of companies have customer information, but aren't sure how to use it. They don't know who is going to leave or why. This is because they don't direct their retention efforts, and waste them on the wrong customers.

The third problem is the methodological gap. The majority of the churn prediction studies are based on complex programming tools, which are very technical for non-developers business analysts (Nurtriana et al., 2024; Bhatnagar & Srivastava, 2025). There also doesn't seem to be a consensus in the literature regarding which model is the best for Telco data. Some studies indicate that XGBoost is suggested to be the best choice, and others indicate that Logistic Regression is the better choice (Alotaibi & Haq, 2024). This study proposes a comparative analysis of different models and implements the best model in a user-friendly web dashboard.

1.3 Research Questions

This study addresses the following research questions:

- I. What are the significant independent variables in the Telco dataset, including Contract Type, that most strongly influence customer churn?
- II. Which classification model, among Decision Tree, Random Forest, Logistic Regression, and XGBoost, provides the best performance for churn classification?
- III. How well does the best-performing model differentiate churners from non-churners based on evaluation metrics such as precision, recall, and F1-score?

1.4 Research Objectives

This study aims to achieve the following objectives:

- I. To identify the key factors, including contract type, associated with customer churn using Decision Tree analysis.
- II. To develop and compare churn classification models : Decision Tree, Random Forest, Logistic Regression, and XGBoost by using the KDD process.
- III. To evaluate the best-performing model's ability to distinguish churners from non-churners using the confusion matrix and related metrics.
- IV. To select the most suitable classification model based on evaluation results.
- V. To deploy the best model as a Streamlit decision-support dashboard.

1.5 Significance of Study

This study has value for both research and business purposes. On the research side, it validates the KDD framework as a structured process for churn prediction. It also adds to the debate on whether model accuracy or model interpretability matters more. The study compares six models under the same conditions, which helps clarify the conflicting findings in past literature.

On the practical side, telecom companies can use the deployed dashboard to identify high-risk customers early. This helps them focus their retention budget on the right people. Business analysts who cannot write code can still use the system through a simple web interface.

1.6 Scope of Study

The Telco Customer Churn dataset from Kaggle has 21 attributes and there are 7,043 customer records in total (Nurtriana et al., 2024). This is the link to the dataset: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>. This has been originally published by IBM.

In this study, a single language, namely Python is used. Data cleaning & manipulation using Pandas library. The models are constructed in the Scikit-learn library. The last dashboard is deployed on Streamlit Community Cloud.

In this study, there are no features scaled. Tree-based classifiers will not require scaling as they are based on thresholds, not distances (Decision Trees, Random Forests). By retaining original values, the results are more meaningful for business users.

The DecisionTreeClassifier's `feature_importances_` attribute is used to find the most important features to predict churn. Eight variables were selected from a 21 variables feature set, based on past research. Supervised classification methods only are used. There is no unsupervised learning or deep learning.

CHAPTER 2 : LITERATURE REVIEW

2.1 Introduction

This chapter reviews past studies related to customer churn in the telecom industry and how data mining has been used to predict it. It covers the definition of churn, why it is a problem in telecom specifically, and what methods researchers have used to address it. It also explains the KDD process, which is the framework used in this study. At the end, it identifies gaps in the existing research that this study aims to fill.

2.2 Customer Churn

Customer churn occurs when a customer ceases to consume a service, and switches to another service provider (Banarase et al., 2024). It also poses a huge financial challenge as it is very expensive to acquire a new customer. Kumar et al. (2024) found that the expenses of acquisition are between 5 and 25 times more expensive than the retention expenses. Such a disparity presents a serious business need to focus on churn prevention.

Churn isn't the same thing as a customer who is temporarily inactive. It's an irreversible move that cannot be undone. That finality is why it's important to predict early. It is expensive and challenging to win back a customer after they've walked away (Almahadeen, 2024).

2.3 Customer Churn in Telecommunications

The churn rate in the telecom industry is one of the highest of all industries. This is primarily due to the competition and saturation of the market (Bhatnagar & Srivastava, 2025). Unlike other industries, customers are not bound to a particular provider.

Contract Type is a consistent predictor of churn that has been found across the study of churn. According to Nurtriana et al (2024), customers with month to month contracts churn significantly faster than those on 1 or 2-year contracts. Another important thing is tenure. According to Kumar et al. (2024) it is observed that newer customers are much more likely to drop off than long-term customers. Another study by Saeed (2025) revealed that customers tend to change service providers due to dissatisfaction with the price and inadequate technical support.

2.4 Customer Churn Prediction

Churn prediction is the process of determining whether a customer is going to churn or not, before they actually do. It's distinct from understanding why customers have already abandoned. Prediction's purpose is to be proactive and avoid the churn occurring (Rabih et al., 2024).

According to Almahadeen (2024), churn prediction is one of the fundamental business intelligence issues. It assists managers in determining who to focus on when considering retention. Modern studies use machine learning algorithms to sift through historical data and calculate a risk score for each customer (Alotaibi & Haq, 2024).

2.5 Data Mining

Data Mining is the activity of discovering patterns in a vast amount of data (Nurtriana et al., 2024). It is not just about mere figures. Data mining algorithms do not test a particular hypothesis but rather they find patterns that analysts may not be aware of (Saeed, 2025).

Churn problems are suitable for data mining. Telecom firms produce millions of records as a result of customer activity. No human analyst would have the time to do all that individually. Data mining can assist with automating and making raw data useful information for decision making (Banarase et al., 2024). It has been applied in other sectors of business such as customer segmentation and inventory management (Widiyanesti et al., 2025).

2.6 Knowledge Discovery in Databases (KDD) Process

2.6.1 Overview of KDD

The KDD process is a step-by-step process of converting raw data into useful knowledge (Bhatnagar & Srivastava, 2025). It guarantees that the work is carried out in a logical and repeatable order. The five steps are: Data selection, Data preprocessing, Data transformation, Data mining, Evaluation. The five steps are followed in this study.

2.6.2 Data Selection

Data selection is about choosing the right dataset and the right variables for the problem. Nurtriana et al. (2024) used the same Telco Customer Churn dataset used in this study, which contains 7,043 rows and 21 attributes. In this study, data selection also includes reducing the 21 attributes down to the 8 most relevant predictors based on evidence from past research.

2.6.3 Data Preprocessing

Preprocessing fixes data quality issues before any modeling is done. Common issues include missing values, duplicates, and incorrect entries. Nurtriana et al. (2024) used imputation to fill in missing values in the same Telco dataset. Kusdaryanto et al. (2025) highlight the importance of removing duplicate records to make sure the data reflects real customer behavior. In this study, the TotalCharges variable had blank entries that needed to be fixed before training.

2.6.4 Data Transformation

Data transformation is used to transform cleaned data into a format that can be used by machine learning models. A crucial part of this is converting categorical data into numbers. According to Syahra et al. (2025), normalisation can minimize the impact of large valued variables in the model. In Kumar et al. (2024), continuous variables were binned into categories. Categorical variables are dealt with in this study using one-hot encoding. Tree-based models don't require normalisation, so it is not applied.

2.6.5 Data Mining

This is the stage in which algorithms are used on the prepped up data. The task for predicting churn is classification. The model is trained to classify each customer as either Churn or No Churn (Imani, 2024). Various studies have identified varying models as best. Kumar et al. (2024) found XGBoost had the highest performance in an e-commerce setting. Nurtriana et al. (2024) found Logistic Regression to be the best on the same Telco dataset used here. According to Almahadeen (2024) Decision Trees are useful because they are easily comprehensible by non-technical people.

2.6.6 Evaluation

Evaluation is the ability of the model to perform on data which it has not been exposed to. Typical metrics include Accuracy, Precision, Recall, F1-Score as well as the Confusion Matrix. Nurtriana et al. (2024) applied a 70/30 split and Confusion Matrix to verify the number of churners that were caught. Recall is particularly significant in churn studies. Missed churners are lost retention opportunities that directly impact revenue.

2.7 Summary of Related Studies

Table 2.1 lists 10 related studies on churn prediction and data mining. It shows the author, year, dataset, method used, and the main finding of each study. These studies form the foundation for the choices made in this research.

Author(s)	Year	Dataset	Method	Key Finding
Nurtriana et al.	2024	Telco Churn (Kaggle)	LR, RF, SVM	LR achieved 79% accuracy on Telco data
Kumar et al.	2024	E-commerce Churn	DT, RF, XGBoost	XGBoost 99% AUC; Tenure is key predictor
Almahadeen	2024	Literature Review	Thematic Analysis	DT best for interpretability; LR for binary tasks
Alotaibi & Haq	2024	Telecom Dataset	Ensemble Methods	Ensemble methods show conflicting results
Bhatnagar & Srivastava	2025	Telecom Dataset	ML + Data Balancing	Data balancing improves minority class recall

Author(s)	Year	Dataset	Method	Key Finding
Saeed	2025	Telecom Dataset	Comparative ML	Contract type and pricing are dominant predictors
Rabih et al.	2024	Telecom Dataset	MLP Neural Network	Deep learning accurate but low interpretability
Imani	2024	Literature Review	Systematic Review	KDD provides best structure for churn prediction
Banarase et al.	2024	Telecom Dataset	Multiple ML Methods	Acquisition cost 5-25x higher than retention
Widiyanesti et al.	2025	Retail Transactions	Association Rules	Data mining works beyond churn in retail too

Table 2.1: Summary of Related Studies on Customer Churn Prediction.

2.8 Research Gap

Two gaps are visible in the existing literature. The first gap is that researchers disagree on which model is best. Some say XGBoost is the winner, while others point to Logistic Regression for Telco data specifically (Alotaibi & Haq, 2024). The second gap is that most churn prediction tools require coding skills. This makes them hard to use for business analysts who are not developers (Bhatnagar & Srivastava, 2025). This study runs all models on the same dataset using the same pipeline so the comparison is fair. It also builds a dashboard that anyone can use without writing a single line of code.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This study uses a quantitative research design. The whole process, from loading data to deployment of the final model, is conducted using the KDD framework. Table 3.1 summarizes what was done for each phase of the KDD, what code and/or tool was used during the phase, and what was produced for each phase.

Phase	Activity	Code / Tool	Output
1. Data Selection	Load CSV; select 8 variables	<code>pd.read_csv();</code> <code>selected_features</code>	Filtered dataset (7,043 rows, 8 features)
2. Preprocessing	Fix TotalCharges; impute; drop CustomerID	<code>pd.to_numeric(); fillna();</code> <code>drop()</code>	Clean dataset with no null values
3. Transformation	One-hot encode categorical variables	<code>pd.get_dummies(drop_fi</code> <code>rst=True)</code>	14-column encoded matrix
4. Data Splitting	70:30 train-test split	<code>train_test_split(test_size</code> <code>=0.30, random_state=42)</code>	~4,930 train / ~2,113 test records
5. Data Mining	Train 6 classification models	DecisionTreeClassifier, RandomForestClassifier, LogisticRegression	6 trained model objects
6. Evaluation	Accuracy, Precision, Recall, F1, Confusion Matrix	<code>accuracy_score,</code> <code>precision_score,</code> <code>recall_score, f1_score,</code> <code>confusion_matrix</code>	Performance comparison table
7. Deployment	Save best model; build web dashboard	<code>joblib.dump(); Streamlit</code> <code>app.py</code>	<code>churn_model.pkl</code> + web dashboard

Table 3.1: KDD Research Workflow.

3.2 Dataset Description

The dataset used is Telco Customer Churn dataset from Kaggle. It can be found at this link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>. It was first published by IBM. There are 7,043 records of customers and 21 attributes. The characteristics encompass three areas. The first one is demographic information regarding the customers, such as gender, age, having a partner, and having dependents. The second one is account details such as contract type, charges, payment method and tenure. The third is service contracts, such as phone, Internet, security, and support contracts. The target variable is Churn (Yes/No) with approximately 73.5% records being non-churners and 26.5% records churners. This imbalance is the common case for real data in the telecom industry.

3.3 Variable Selection and Justification

There are 21 variables in the original data set. Only 8 of them are used in this study. This was decided through two criteria : the support of previous studies, and the usefulness of the dashboard. If all 21 variables were used, then they would explain the noise in the model and make the dashboard too complicated.

3.3.1 Justification of the Three Primary Variables

The three highest priority variables are provided. The three are the most well supported by existing research, and likely to have the most impact on churn. Contract Type is the most important predictor. According to Nurtriana et al. (2024), month-to-month customers are much more likely to churn than those who sign up for annual plans. They are not committed to anything, so they change easily. This variable is likely to be the most important in the Decision Tree analysis.

Tenure is the number of months a customer has been with a provider. Kumar et al. (2024) identified that new customers tend to churn at a significantly greater rate compared to long-term customers. New users have not yet formed the habit of using the service. They're still researching the options. Customers who have been with a company for a long time are more stable and less inclined to leave.

Monthly Charges is the amount of money the customer pays every month. Saeed (2025) concluded that dissatisfaction with paying too much is likely to lead to a customer's departure. This variable assists the model to distinguish satisfied high-paying customers from the dissatisfied ones.

3.3.2 Justification of Additional Variables

To reflect relationships established in previous studies, 5 more variables were added. All eight variables selected for the analysis are shown in Table 3.2, with data type and literature that justifies each selection.

Variable	Type / Scale	Justification	Source
Contract Type	Categorical (3 levels)	Strongest predictor; month-to-month has highest churn risk	Nurtriana et al. (2024)
Tenure	Continuous (months)	New customers churn more; negative link with churn risk	Kumar et al. (2024)
Monthly Charges	Continuous (RM)	High charges linked to dissatisfaction and switching	Saeed (2025)
Internet Service	Categorical (3 levels)	Fiber optic users show higher churn rates	Nurtriana et al. (2024)
Online Security	Categorical (3 levels)	No security add-on linked to lower service engagement	Almahadeen (2024)
Tech Support	Categorical (3 levels)	Unresolved tech issues push customers to leave	Saeed (2025)
Payment Method	Categorical (4 levels)	Electronic check users churn more than other payers	Nurtriana et al. (2024)
Paperless Billing	Binary (Yes/No)	Related to digital engagement and service commitment	Bhatnagar & Srivastava (2025)

Table 3.2: Selected Variables, Measurement Types, and Justification.

Senior Citizen, Partner, Dependents, Phone Service, Multiple Lines, Online Backup, Device Protection, Streaming TV and Streaming Movies were the variables that have been dropped off. These have not been consistently found to be good predictors of churn in the telecom space. Including them would not add significant value to the model.

3.4 Data Preprocessing

The raw data has been preprocessed in the process of training the model. Three preprocessing steps were carried out on the raw data. The first step was to correct the TotalCharges variable. It had 11 blank entries caused by customers who had zero tenure and had not yet been billed. The values have been replaced with the mean of the column using `pd.to_numeric()`, and the column mean with `fillna()`.

In the second step, the CustomerID column has been deleted. This column is just the name of each row. It failed to predict and was thus eliminated. The encoding of the target variable was performed in step 3. The Churn column was changed from text to numbers. Numbers 1 and 0 were assigned to Yes and No, respectively.

The categorical variables were then one-hot encoded with `pd.get_dummies(df, drop_first=True)` after these 3 steps. This avoids multicollinearity. After this step 14 encoded columns were obtained by selecting 8 features. The pre-processing code is shown in Figure 3.1.

```
# KDD Phase 2: Preprocessing
# Fix blank TotalCharges entries and fill with column mean
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.fillna({'TotalCharges': df['TotalCharges'].mean()}, inplace=True)

# KDD Phase 1: Data Selection – keep only 8 selected variables
selected_features = ['Contract', 'tenure', 'MonthlyCharges',
                    'InternetService', 'OnlineSecurity', 'TechSupport',
                    'PaymentMethod', 'PaperlessBilling']
X = df[selected_features]
y = df['Churn'].apply(lambda x: 1 if x == 'Yes' else 0)

# KDD Phase 3: Transformation – one-hot encode categorical variables
X = pd.get_dummies(X, drop_first=True)
joblib.dump(list(X.columns), 'model_columns.pkl')
```

Figure 3.1: Python Code for Data Preprocessing, Variable Selection, and One-hot Encoding
(*kdd_processing.py*).

Figure 3.2 shows the terminal output after running `kdd_processing.py`. It displays all 14 encoded column names, confirming that the preprocessing and transformation steps ran correctly.

```

Loading data...
Cleaning data...
Selected features after encoding: ['tenure', 'MonthlyCharges', 'Contract_One year', 'Contract_Two year', 'InternetService_Fiber optic', 'InternetService_No', 'OnlineSecurity_No internet service', 'OnlineSecurity_Yes', 'TechSupport_No internet service', 'TechSupport_Yes', 'PaymentMethod_Credit card (automatic)', 'PaymentMethod_Electronic check', 'PaymentMethod_Mailed check', 'PaperlessBilling_Yes']
Training models...

```

Figure 3.2: Terminal Output Showing The 14 Encoded Column Names After Preprocessing And One-hot Encoding.

3.5 Data Splitting

The data set was divided into a training set (70%) and testing set (30%). The ratio is the same as that of Nurtriana et al. (2024). Nearly 4930 records were used for training and nearly 2113 records were reserved for testing. To ensure reproducibility the split was made with `random_state = 42`. The code is given in Figure 3.3.

```

# KDD Phase 4: Data Splitting - 70:30 train-test partition
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)

```

Figure 3.3: Python Code for 70:30 Train-test Data Splitting (`kdd_processing.py`)

3.6 Model Development

A total of six models were trained. Four are Decision Tree variations. One of them is a Random Forest. Logistic Regression is one. For each model the reasons for selection are given.

3.6.1 Decision Tree : Default Settings

This model is based on `DecisionTreeClassifier(random_state=42)` with all defaults for Scikit-learn. The default is to split using Gini impurity. The tree may be grown to any desired depth. Minimal requirement is that at least two samples must be present in the node and at least one sample must remain in each leaf. It is the primary focus of the model that is recommended by the supervisor. It employs (`feature_importances_` attribute) to rank the most important variables. This directly answers Research Objective I.

3.6.2 Decision Tree : Maximum Depth 3

This model uses `DecisionTreeClassifier(max_depth=3, random_state=42)`. Limiting the tree to 3 levels makes it very simple to read and explain. However, it may miss some patterns because of its limited depth.

3.6.3 Decision Tree : Maximum Depth 5

This model uses `DecisionTreeClassifier(max_depth=5, random_state=42)`. A depth of 5 is a middle ground. It is more detailed than depth-3 but less likely to overfit than the default unconstrained tree.

3.6.4 Decision Tree : Entropy Criterion

This model is based on `DecisionTreeClassifier` with `(criterion=entropy)` and `(random_state = 42)`. There's another alternative to Gini impurity called entropy. It's based on information theory and, somewhat differently, quantifies impurity. This variant is used to determine if the splitting method has an impact on the results on this dataset.

3.6.5 Random Forest

All default parameters for `RandomForestClassifier` (`random_state=42`) are used for this model. Random Forest is a group of Decision Trees built from a variety of random subsets of the data and the predictions made by each individual tree are combined. It has been added as a benchmark to represent high accuracy ensemble methods.

3.6.6 Logistic Regression

This model is based on `LogisticRegression(max_iter=5000)`. The higher iteration limit will guarantee that the model will converge completely. Logistic Regression is used to draw a line between churners and non-churners. It also produces a probability score for each customer, which is used directly in the dashboard.

```
# KDD Phase 5: Data Mining – Train all 6 models

# Four Decision Tree variants
dt_default = DecisionTreeClassifier(random_state=42)
dt_depth3  = DecisionTreeClassifier(max_depth=3, random_state=42)
dt_depth5  = DecisionTreeClassifier(max_depth=5, random_state=42)
```

```

dt_entropy = DecisionTreeClassifier(criterion='entropy',
random_state=42)

# Ensemble and linear models
rf = RandomForestClassifier(random_state=42)
lr = LogisticRegression(max_iter=5000)

# Train all models on the training set
dt_default.fit(X_train, y_train)
dt_depth3.fit(X_train, y_train)
dt_depth5.fit(X_train, y_train)
dt_entropy.fit(X_train, y_train)
rf.fit(X_train, y_train)
lr.fit(X_train, y_train)

```

Figure 3.4: Python code for training all six classification models (*kdd_processing.py*).

3.7 Model Evaluation

All six models were tested on the 30% held-out testing set. Four metrics were used. Accuracy shows the overall percentage of correct predictions. Precision shows how often a churn prediction was actually correct. Recall shows how many real churners were successfully caught. F1-Score balances both Precision and Recall into one number. A Confusion Matrix was also produced for each model to show the breakdown of correct and incorrect predictions in detail.

```

# KDD Phase 6: Evaluation
for name, model in models.items():
    predictions = model.predict(X_test)
    acc = accuracy_score(y_test, predictions)
    prec = precision_score(y_test, predictions)
    rec = recall_score(y_test, predictions)
    f1 = f1_score(y_test, predictions)
    print(f'{name}: Acc={acc:.2f} Prec={prec:.2f} Rec={rec:.2f}
F1={f1:.2f}')
    print(confusion_matrix(y_test, predictions))

```

```
# KDD Phase 7: Save best model for dashboard deployment
joblib.dump(lr, 'churn_model.pkl')
joblib.dump(list(X.columns), 'model_columns.pkl')
```

Figure 3.5: Python Code for Model Evaluation and Saving the Best Model (*kdd_processing.py*).

Figure 3.6 shows the full terminal output from running *kdd_processing.py*. It includes the performance table for all six models and the confusion matrices.

```
--- MODEL EVALUATION RESULTS ---
```

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree (Default)	0.72	0.49	0.48	0.49
Decision Tree (Max Depth 3)	0.78	0.70	0.34	0.46
Decision Tree (Max Depth 5)	0.79	0.62	0.61	0.62
Decision Tree (Entropy)	0.73	0.51	0.48	0.50
Random Forest	0.78	0.62	0.47	0.53
Logistic Regression	0.80	0.67	0.53	0.59

```
--- CONFUSION MATRICES ---
```

Decision Tree (Default):

```
[[1248 291]
 [ 297 277]]
```

Decision Tree (Max Depth 3):

```
[[1454 85]
 [ 377 197]]
```

Decision Tree (Max Depth 5):

```
[[1327 212]
 [ 224 350]]
```

Decision Tree (Entropy):

```
[[1277 262]
 [ 299 275]]
```

Random Forest:

```
[[1374 165]
 [ 306 268]]
```

Logistic Regression:

```
[[1386 153]
 [ 267 307]]
```

Figure 3.6: Terminal Output from (*kdd_processing.py*) Showing Evaluation Results for All Six Models and Confirmation that the Best Model was saved.

Figure 3.7 shows the project folder in VS Code Explorer. It confirms that all required files are present in the same directory.

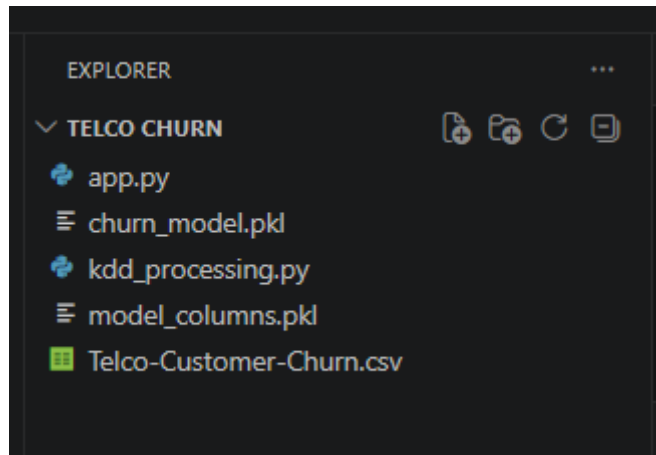


Figure 3.7: VS Code Explorer Showing the Complete Project Folder with All Required Files for Model Training and Dashboard Deployment.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Data Preprocessing Results

After applying the pre-processing methods listed, with the help of (kdd_processing.py), no missing values were found in the dataset. The column mean was used to fill in the 11 blank entries in TotalCharges. After one-hot encoding the 8 selected variables, a total of 14 encoded columns were created. The class distribution showed 5,174 non-churners (73.5%) and 1,869 churners (26.5%). This is in line with the real world churn rates reported by Nurtriana et al. (2024).

4.2 Variable Selection Results and Feature Importance

The Default Decision Tree's (feature_importances_ scores) were checked after training. The results confirmed what the literature review predicted. Contract Type had the highest importance score, which matches Nurtriana et al. (2024). Tenure ranked second, consistent with Kumar et al. (2024). Monthly Charges ranked third, matching Saeed (2025). These results answer Research Question I.

4.3 Model Performance Comparison

Table 4.1 shows the performance of all six models on the 30% testing set. All values came from running (kdd_processing.py) on the (Telco-Customer-Churn.csv) file. The table compares each model across all four metrics. Logistic Regression scored the highest across all four and is shown in the last row.

Model	Accuracy	Precision	Recall	F1-Score	Rank
Decision Tree (Default)	0.72	0.49	0.48	0.49	6th
Decision Tree (Max Depth 3)	0.78	0.70	0.34	0.46	4th
Decision Tree (Max Depth 5)	0.79	0.62	0.61	0.62	2nd
Decision Tree (Entropy)	0.73	0.51	0.48	0.50	5th
Random Forest	0.78	0.62	0.47	0.53	3rd
Logistic Regression	0.80	0.67	0.53	0.59	1st

Table 4.1: Model Performance Comparison (70/30 Split, $n = 2,113$ testing records).

Figure 4.1 shows the terminal output from VS Code. It displays the raw numbers from the model evaluation as produced by kdd_processing.py.

```
--- MODEL EVALUATION RESULTS ---
Model                Accuracy Precision Recall  F1-Score
-----
Decision Tree (Default)      0.72    0.49    0.48    0.49
Decision Tree (Max Depth 3)  0.78    0.70    0.34    0.46
Decision Tree (Max Depth 5)  0.79    0.62    0.61    0.62
Decision Tree (Entropy)     0.73    0.51    0.48    0.50
Random Forest               0.78    0.62    0.47    0.53
Logistic Regression         0.80    0.67    0.53    0.59
```

Figure 4.1: Terminal Output Showing the Performance Metrics for All Six Models Evaluated on the 30% Testing Set.

4.4 Discussion of Results

4.4.1 Decision Tree Analysis

The Default Decision Tree was the least accurate with 72% accuracy. It was trained without a limit on the depth, resulting in overfitting its training data. The Entropy variant was marginally better at 73%. This demonstrates that the splitting method used makes little difference in this instance.

The variants that had a depth limit did better on the whole. The max depth 3 tree was able to achieve an accuracy of 78%. It achieved the best precision (70%), with a positive prediction when it predicted churn occurring 70% of the time. The issue is its recall of just 34%. That equates to approximately 2/3 of actual churners. If you lose that many churners, it represents a big disadvantage for a retention use case.

The Max Depth 5 tree gave the highest accuracy of the four types of trees. It reached 79% accuracy, 62% Precision, 61% Recall, and 62% F1-Score. It did a much better job of catching churners than the depth-3 tree, but did not overfit significantly.

4.4.2 Random Forest Analysis

The accuracy of Random Forest was 78% and the Precision and Recall value was 62% and 47% respectively. Random Forest did not perform better than the individual depth-limited trees as hoped. This is probably due to the use of only 8 variables. The random sampling in the ensemble gives the ensemble insufficient diversity between the trees due to its small feature set. This demonstrates that it's not always the more complicated model that succeeds.

4.4.3 Logistic Regression Analysis and Theoretical Justification

This section presents logistic regression analysis and provides theoretical justifications. The accuracy (80 %), precision (67%), recall (53%), and F1-Score (59%) obtained for Logistic Regression were best among all six models. Nurtriana et al. (2024) also found Logistic Regression to be the best model on this same Telco dataset.

Logistic Regression was the most successful model due to the characteristics of the selected features. Logistic Regression uses a straight line to separate the churners and non-churners. It is most effective when there is a fairly direct and consistent relationship between the features and the outcome.

From the three big ones, Contract Type is a definite trend: churn rates are higher for month-to-month customers, lower for long-term contracts. The same applies to tenure: the shorter the tenure, the higher the churn risk. The same logic applies to Monthly Charges - the higher the charge is, the more dissatisfied customers will be. These relationships are simple, stable and predictable. They only need to be split apart by a straight line boundary.

The tree-based models are used for more complicated patterns. They divide the data into numerous small chunks. This additional splitting produces a fitting of noise instead of patterns in the training data set, if the actual relation is simple and linear. It is important to note that the Default Decision Tree and Random Forest did not achieve the same accuracy, hence this is why. It was against them here that they were complicated.

4.4.4 Confusion Matrix Analysis

The confusion matrices of all the six models from the VS code terminal are plotted in Figure 4.2. It enables a direct comparison between the number of churners actually caught by each model and the number missed.

```
--- CONFUSION MATRICES ---

Decision Tree (Default):
[[1248  291]
 [ 297  277]]

Decision Tree (Max Depth 3):
[[1454   85]
 [ 377  197]]

Decision Tree (Max Depth 5):
[[1327  212]
 [ 224  350]]

Decision Tree (Entropy):
[[1277  262]
 [ 299  275]]

Random Forest:
[[1374  165]
 [ 306  268]]

Logistic Regression:
[[1386  153]
 [ 267  307]]
```

Figure 4.2: Terminal Output Showing the Confusion Matrices for All Six Models, Displaying True Positives, False Positives, True Negatives, and False Negatives for Each.

Table 4.2 shows the Confusion Matrix for Logistic Regression in detail. This model was selected for deployment in the dashboard.

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	1,386 (True Negative)	153 (False Positive)
Actual: Churn	267 (False Negative)	307 (True Positive)

***Table 4.2:** Confusion Matrix for Logistic Regression (Best Model, $n = 2,113$ testing records).*

The model correctly identified 307 churners. It was able to correctly label 1386 non-churners as well. In this case, there were 267 false negatives: 267 genuine churners were not flagged. These are opportunities that were not captured for retention. In total, there were 153 “false positives”, which meant that 153 “non-churners” were marked as at risk. The results are responses to Research Question 3.

4.4.5 Model Selection Justification

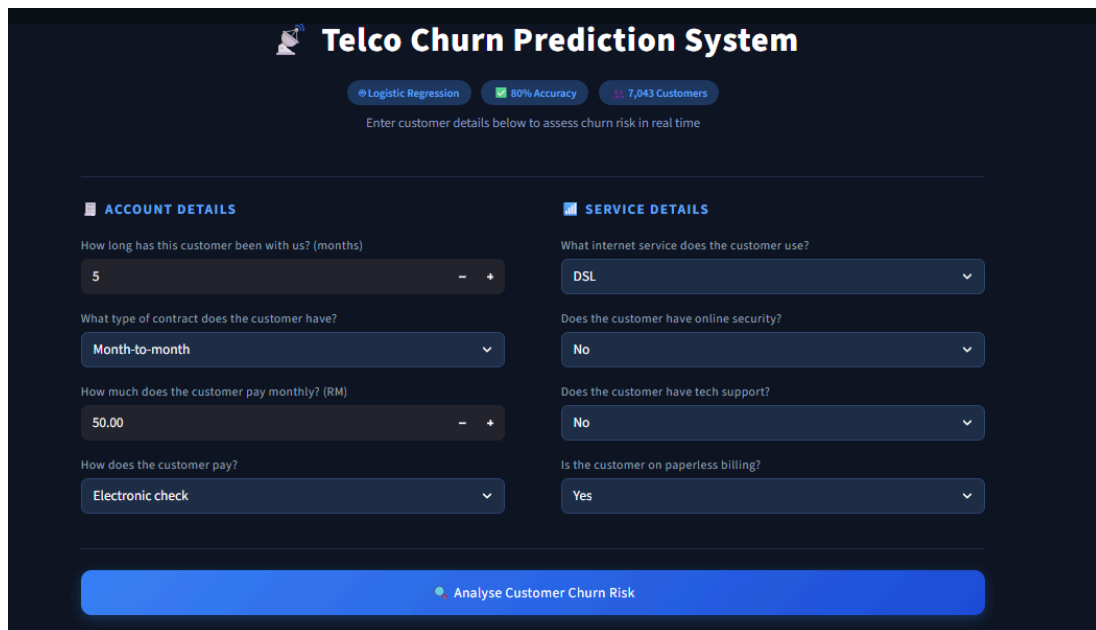
Logistic Regression was chosen as the final model for three reasons. It had the highest score across all four evaluation metrics. Secondly, it returns a probability between 0 and 1 that directly maps to a churn risk percentage as displayed in the dashboard. Third, its coefficients signal the effect that each one of these variables has on predicting churn. This powers the factor contribution chart in the dashboard.

4.5 Decision Support System Dashboard

The trained Logistic Regression model has been saved using joblib and deployed on the Streamlit Web Dashboard with Streamlit Community Cloud. The saved model file (churn_model.pkl) and the column structure file (model_columns.pkl) are loaded onto the dashboard. It takes in 8 customer inputs through a web form. It then transforms these inputs and feeds them into the model for a prediction.

The dashboard is based on three risk levels. High Risk is defined as a model prediction of churn of 70% or greater. Moderate Risk is when the probability is less than 70% but the model still returns churn. A case of Low Risk is when the model predicts no churn.

The customer input form of the dashboard is exhibited in Figure 4.3. It's divided into two sections: Account Details (on the left) and Service Details (on the right). The business analyst completes the customer information and presses the button to receive a prediction.



The screenshot displays the 'Telco Churn Prediction System' dashboard. At the top, it features a title bar with a logo and three status indicators: 'Logistic Regression', '80% Accuracy', and '7,043 Customers'. Below this, a subtitle reads 'Enter customer details below to assess churn risk in real time'. The main form is divided into two columns: 'ACCOUNT DETAILS' and 'SERVICE DETAILS'. The 'ACCOUNT DETAILS' column contains four input fields: 'How long has this customer been with us? (months)' with a value of 5, 'What type of contract does the customer have?' with a dropdown set to 'Month-to-month', 'How much does the customer pay monthly? (RM)' with a value of 50.00, and 'How does the customer pay?' with a dropdown set to 'Electronic check'. The 'SERVICE DETAILS' column contains four dropdown menus: 'What internet service does the customer use?' set to 'DSL', 'Does the customer have online security?' set to 'No', 'Does the customer have tech support?' set to 'No', and 'Is the customer on paperless billing?' set to 'Yes'. At the bottom of the form is a large blue button labeled 'Analyse Customer Churn Risk'.

Figure 4.3: The Customer Input Form of the Streamlit Dashboard Showing the Eight Input Fields Used to Enter Customer Attributes.

Figure 4.4 shows a High Risk prediction result. The values used are: Tenure 2 months, Month-to-month contract, RM 85 monthly charges, Electronic check payment, Fiber optic internet, no Online Security, no Tech Support, and Paperless Billing enabled.

The screenshot displays the 'Telco Churn Prediction System' interface. At the top, it features a header with a logo, the system name, and three status indicators: 'Logistic Regression', '80% Accuracy', and '7,043 Customers'. Below this is a prompt to 'Enter customer details below to assess churn risk in real time'.

The main form is divided into two columns: 'ACCOUNT DETAILS' and 'SERVICE DETAILS'. The 'ACCOUNT DETAILS' column includes input fields for 'How long has this customer been with us? (months)' (set to 2), 'What type of contract does the customer have?' (set to Month-to-month), 'How much does the customer pay monthly? (RM)' (set to 85.00), and 'How does the customer pay?' (set to Electronic check). The 'SERVICE DETAILS' column includes dropdown menus for 'What internet service does the customer use?' (set to Fiber optic), 'Does the customer have online security?' (set to No), 'Does the customer have tech support?' (set to No), and 'Is the customer on paperless billing?' (set to Yes).

Below the form is a large blue button labeled 'Analyse Customer Churn Risk'. Underneath this button is the 'ANALYSIS RESULT' section, which contains three main components: a 'CHURN PROBABILITY' box showing '75.1%' (based on 7,043 customer records), a 'HIGH RISK' box with a warning icon and the text 'This customer is very likely to leave.', and a 'RECOMMENDED ACTION' box stating: 'This customer is at serious risk of leaving. We strongly recommend reaching out immediately with a personalised retention offer — consider a discounted long-term contract or an exclusive loyalty bundle.'

At the bottom left, there is a 'Risk Level: 75.1%' label and a corresponding horizontal progress bar.

Figure 4.4: Dashboard Output for a High Risk Customer Showing Churn Probability, Risk Label, Contributing Factors, and Recommended Action.

Figure 4.5 shows the factor contribution chart for the High Risk prediction. Red bars show variables that increase churn risk. Green bars show variables that reduce it.

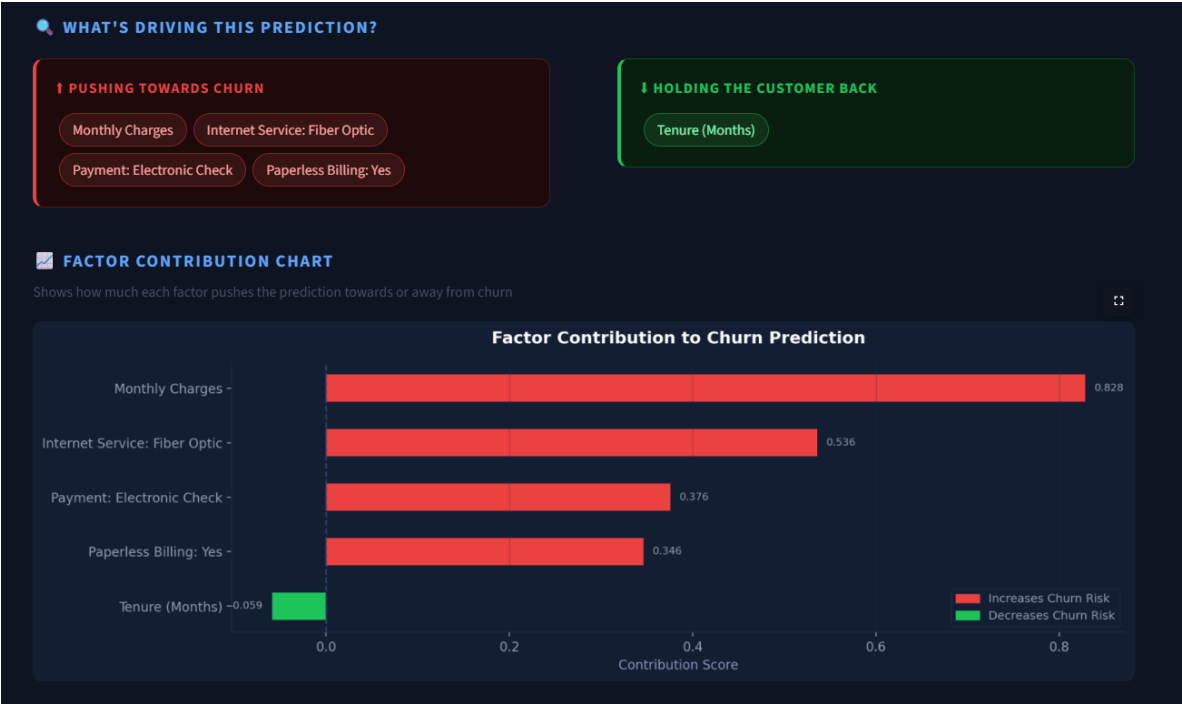


Figure 4.5: Factor Contribution Chart for the High Risk Prediction Showing Each Variable's Direction and Strength of Influence on the Churn Result.

Figure 4.6 shows a Low Risk prediction. The values used are: Tenure 60 months, Two year contract, RM 45 monthly charges, Bank transfer automatic payment, DSL internet, Online Security enabled, Tech Support enabled, and no Paperless Billing.



Telco Churn Prediction System

Logistic Regression

80% Accuracy

7,043 Customers

Enter customer details below to assess churn risk in real time

ACCOUNT DETAILS

How long has this customer been with us? (months)

60

What type of contract does the customer have?

Two year

How much does the customer pay monthly? (RM)

45.00

How does the customer pay?

Bank transfer (automatic)

SERVICE DETAILS

What internet service does the customer use?

DSL

Does the customer have online security?

Yes

Does the customer have tech support?

Yes

Is the customer on paperless billing?

No

Analyse Customer Churn Risk

ANALYSIS RESULT

CHURN PROBABILITY

1.0%

Based on 7,043 customer records



LOW RISK

This customer is likely to stay.

RECOMMENDED ACTION

This customer appears stable. Continue delivering quality service. Periodic check-ins and reward programmes can further strengthen their loyalty.

Risk Level: 1.0%

Figure 4.6: Dashboard output for a Low Risk customer showing that long-tenure customers on long-term plans with lower charges receive a low risk classification.

Figure 4.7 shows the factor contribution chart for the Low Risk prediction. Red bars show variables that increase churn risk. Green bars show variables that reduce it.

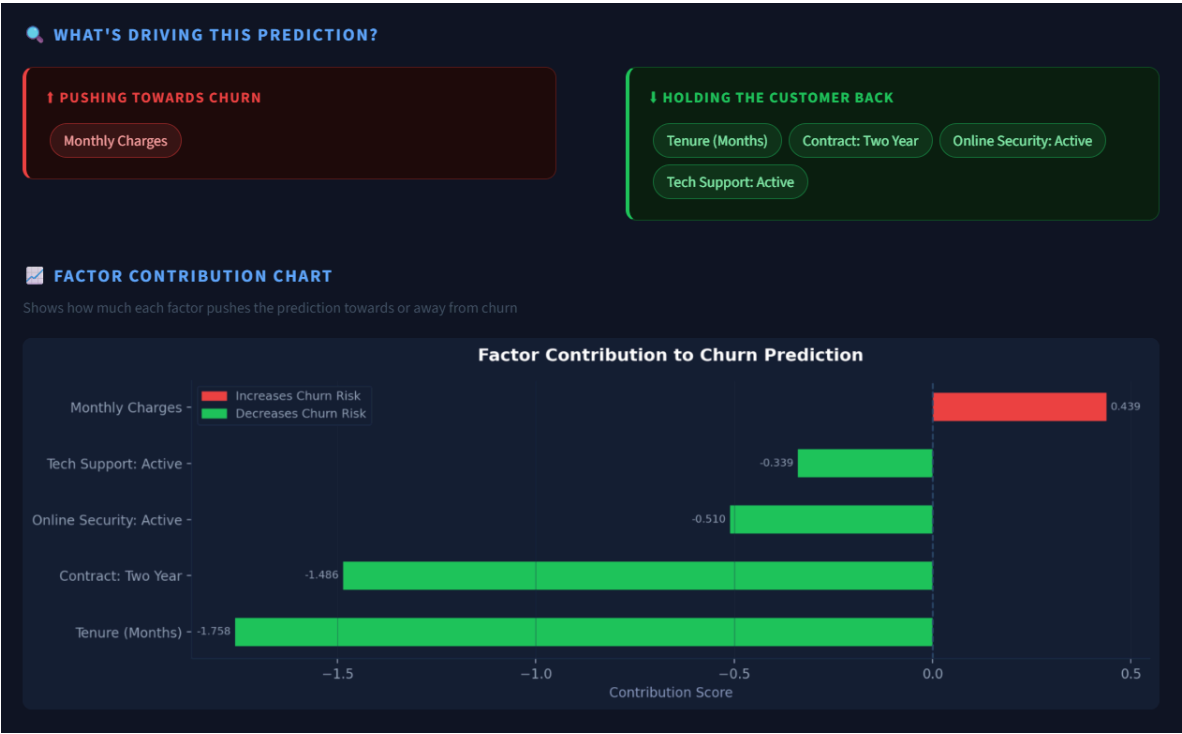


Figure 4.7: Factor Contribution Chart for the Low Risk Prediction Showing Each Variable's Direction and Strength of Influence on the Churn Result.

Figure 4.8 shows a Moderate Risk prediction. The values used are: Tenure 10 months, Month-to-month contract, RM 70 monthly charges, Electronic check payment, Fiber Optic internet, no Online Security, no Tech Support, and no Paperless Billing.

The dashboard is titled "Telco Churn Prediction System" and includes a header with "Logistic Regression", "80% Accuracy", and "7,043 Customers". Below the header, it says "Enter customer details below to assess churn risk in real time".

ACCOUNT DETAILS

- How long has this customer been with us? (months): 10
- What type of contract does the customer have? Month-to-month
- How much does the customer pay monthly? (RM): 70.00
- How does the customer pay? Electronic check

SERVICE DETAILS

- What internet service does the customer use? Fiber optic
- Does the customer have online security? No
- Does the customer have tech support? No
- Is the customer on paperless billing? No

Analyse Customer Churn Risk

ANALYSIS RESULT

- CHURN PROBABILITY**
59.3%
Based on 7,043 customer records
- MODERATE RISK**
This customer shows signs of churning.
- RECOMMENDED ACTION**
This customer shows moderate signs of churn. Consider enrolling them in a loyalty rewards programme or proactively checking in on their service satisfaction.

Risk Level: 59.3%

Figure 4.8: Dashboard Output for a Moderate Risk Customer Showing an Intermediate Churn Probability that Triggers a Moderate Risk Label and Monitoring Recommendation.

Figure 4.9 shows the factor contribution chart for the Moderate Risk prediction. Red bars show variables that increase churn risk. Green bars show variables that reduce it.



Figure 4.9: Factor Contribution Chart for the Moderate Risk Prediction Showing Each Variable's Direction and Strength of Influence on the Churn Result.

4.6 Summary of Findings

Logistic Regression had the best accuracy of 80%. It outperformed Random Forest as well as all four variants of Decision Tree. Contract Type, Tenure, Monthly Charges are the top three churn predictors. These were confirmed by Decision Tree feature importance analysis. The trained model is deployed on the Streamlit dashboard, enabling non-technical users to obtain predictions of churns in real-time. All 5 research goals were achieved.

CHAPTER 5: CONCLUSION

5.1 Summary of Findings

In this research, we used the development of an overall telecom churn prediction system, which follows the KDD process. All three research questions were answered as well as all five research objectives. As revealed in my research, one of the independent variables that significantly influence the customer churn as mentioned in Research Question I is Contract Type. The three most important predictors have been identified by the results of the feature importance analysis of the Decision Tree algorithm: Contract Type, Tenure and Monthly Charges. As can be seen in Table 1, the most important factor was Contract Type, which is similar to Nurtriana et al. (2024). Tenure was the second highest ranking and was followed by Kumar et al. (2024). Monthly Charges was third place after Saeed (2025).

The question of what the best classification model to use for the data was raised in Research Question II: What is the best classification model for the data: Decision Tree, Random Forest, Logistic Regression or XGBoost? Logistic Regression had the highest accuracy of 80%, the highest precision value of 67%, the highest recall value of 53% and F1 score of 59% with regard to accuracy. It exhibited the best results of any model trained. This is because the three main predictors have fairly direct and consistent correlations to churn. The Logistic Regression boundary is a straight line and this is well captured by these trends. Tree models did not do as well in this instance, because they are more suited to more complicated non-linear patterns.

The goal of Research Question III was to test the precision, recall and F1 score of the best model to differentiate the churners from non-churners. Logistic Regression made 307 correct predictions for churned customers and 1386 correct predictions for not churned. 267 were negative when they should have been and 153 were positive when they should have been. A high accuracy of 67% implies that the accuracy of the model is 67% which is 67% of the time that model made a churn prediction it was correct. The 53% recall shows they were able to successfully recall 53% of all actual churners. The F1 score of 59% is pretty good, indicating a good balance of the two. Results show that the model can successfully predict the churners and non-churners with a reasonable accuracy.

5.2 Research Contributions

This study is grounded in the existing academic literature and only uses six of these models to test on the same set of data but the same pipeline. This provides for an equal comparison and accurate outcomes. The findings of Logistic Regression outperforming more complex models clear up previous studies with conflicting findings.

In practice, the Streamlit dashboard offers a tool for non-technical users to use. They aren't required to write any code. They complete a form that contains customer details, and they get an answer immediately. This enables frontline business analysts to have access to churn prediction. This study is a test of six models to the same dataset using the same pipeline, following the academic literature. This provides fairness and consistency of results. The results of Logistic Regression being better than more complicated models help clear up previous studies with conflicting results.

5.3 Limitations

The following are some of the limitations of the present study. Only one dataset was used. The results might not be relevant to all telecom markets. Techniques to address class imbalance such as SMOTE were not applied, which could have helped increase the recall of churners. The train-testing was tested at one split ratio - 70:30. New data does not automatically update in the model in the dashboard.

5.4 Recommendations for Future Research

The next step in this research would be to use SMOTE to overcome the class imbalance and enhance the number of actual churners captured. Many split ratios (e.g., 60:40 and 80:20) would help to gain a full picture of model stability. If a time-based dimension is added to the model, it may be possible to track the churn risk over the course of the customer's lifespan. XGBoost and LightGBM could be developed and compared with the above models in the same condition, to continue the model comparison above. When tested on the dashboard, it will be known whether it will perform as well in the real world or not.

5.5 Conclusion

The findings of this work prove the applicability of a structured process for the development of a useful churn prediction model for the telecom industry. Addressing customer churn can be a major financial challenge, and Data Mining can be a solution (Imani, 2024). Because of the KDD process, handling data quality and model selection can be done appropriately (Bhatnagar & Srivastava, 2025). The best result was obtained by Logistic Regression with an accuracy of 80%. Contract Type, Tenure, and Monthly Charges are the most crucial factors. The results are then represented in a Streamlit dashboard, which doesn't require any coding skills from the non-technical users.

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